

CAN PASTORAL NOMADS RETURN TO THEIR TRADITIONAL LIVELIHOOD OF RAISING LIVESTOCK IN THE REGISTAN DESERT OF SOUTHERN AFGHANISTAN?

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Goal Of This Report

In Afghanistan,¹ there are several million nomadic people known as *kuchis*, a generic term used there mainly by non-nomads to describe pastoral nomads. A severe drought in 1998 caused the displacement of approximately 100,000 *kuchis* from the Registan desert² in southern Afghanistan. Most of these people are now living in temporary settlements between the Arghandab and Helmand Rivers and Registan. Approximately 57,000 are being supported by the UN in camps for internally displaced persons (IDPs) in Maywand and Panjwayi districts of Kandahar Province; the rest are living in unsupported settlements.

The UN is currently seeking strategies to return the *kuchis* to their traditional livelihood of raising livestock in Registan and end UN support of the IDP camps. This also appears to be the desire of most *kuchis*, at least the elders whom we interviewed, although some *kuchis* believe that the nomadic lifestyle is coming to an end (Amani 2003). How can this be achieved and sustained? We start with the premise that the return of *kuchis* to their traditional nomadic lifestyle is possible if carefully planned, seeing that it was done successfully by *kuchis* themselves for many centuries.

Our goal was to examine possibilities for the return of *kuchis* to Registan. We reviewed reports published to date, visited several well clusters in Registan, interviewed *kuchis* from the IDP camps and non-IDP settlements and had discussions with personnel from a number of organisations working with *kuchis*, including the United Nations High Commission on Refugees (UNHCR), United Nations Assistance Mission in Afghanistan (UNAMA), Voluntary Association for Rehabilitation of Afghanistan (VARA) and Central Asian Development Group (CADG). Reports included recent independent studies that described the past and present situation of *kuchis*, the state of vegetation, soils and well clusters and water catchments (*nawars*) in Registan. We were dealing with the post-1998 period; however, there exists a large body of research on pastoral nomads in Afghanistan prior to that date. We did not intend to report on the impact of earlier droughts nor make comparisons with earlier situations. We visited Afghanistan from 21 May to 28 May 2004³.

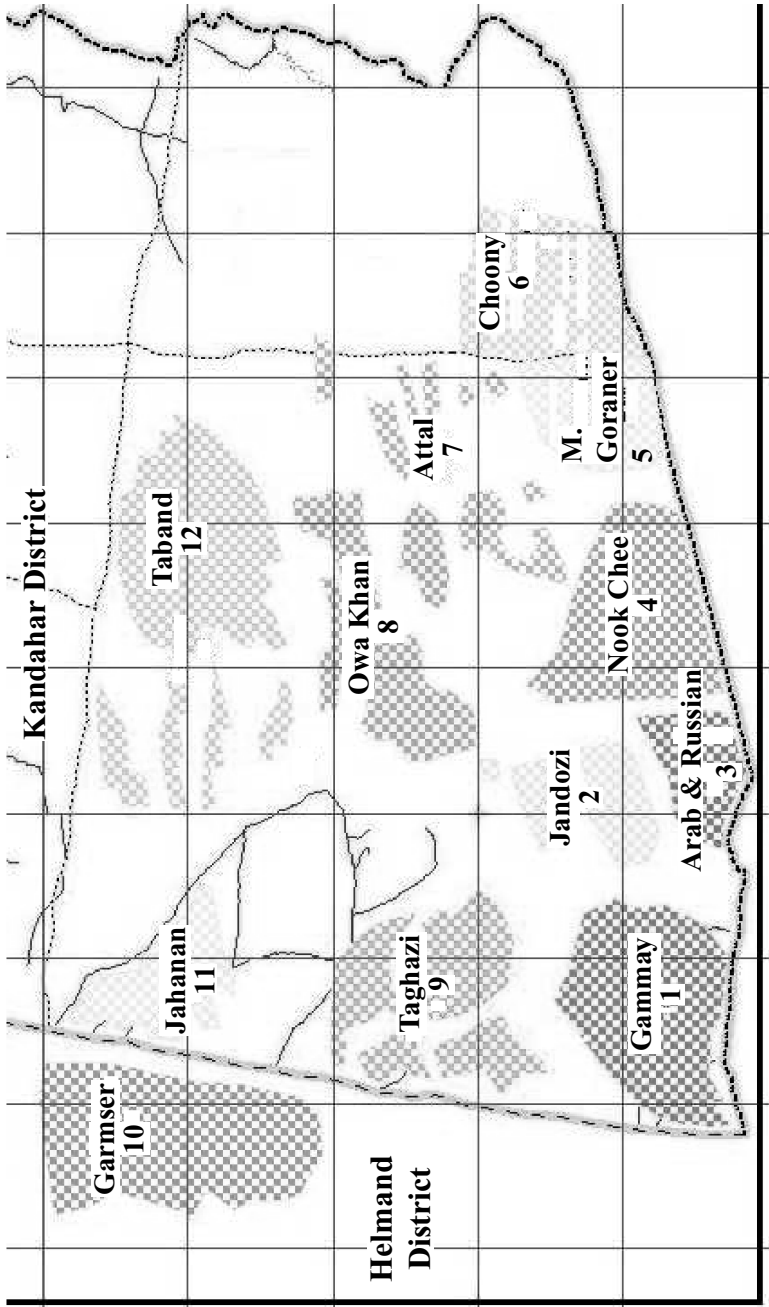


Figure 1: Map Showing 12 Well Clusters in the Registan Desert

Background

Registan is situated in southern Afghanistan, bordered by the sand dunes south of the Arghandab River in the north and by the Pakistan border in the south. It is part of the African-Indian Desert region with a total area of about 24,000 km² that includes part of the Kandahar, Helmand and Nimroz districts. It has a Mediterranean arid climate characterised by low winter rainfall from November to May, high summer air temperatures (above 40°C in August) and a high rate of potential evaporation (about 3000 mm annually). This region is used by nomads and semi-nomads who, until recently, depended solely on the raising of livestock for their livelihood.

Approximately 10 percent (2–2.5 million people) of the population of Afghanistan consists of *kuchis*. They comprise a number of ethnic communities, the Pashtuns being the majority group (UNCHA 2004). In Registan there are only two main ethnic groups: Baluch, comprising about 80 percent, and Pashtuns about 20 percent of the nomadic population. They raise mainly *Baluchi* sheep for meat, milk and wool. In addition, camels are raised for transport, milk and drawing water, goats for milk, hair and meat and donkeys for transport.

Wells and *nawars* are the sources of water for *kuchis* and their animals in Registan. *Nawars* are dug-out depressions that act as collecting reservoirs for run-off water. Only Baluch *kuchis* have wells; Pashtun *kuchis* are dependent on *nawars*. Some Baluch *kuchis* having no access to wells are dependent on *nawars* as well. Livestock in Registan grazes only natural vegetation; apparently, no supplementary feed is offered. According to the *kuchis* whom we interviewed, two methods of livestock raising is practised:

- (1) Grazing animals in Registan during winter/spring months of the year and then migrating to near the Arghandab River or Helmand River areas for the rest of the year. This was practised by all Pashtun *kuchis* as the *nawars* dried out by the summer, and by some Baluch *kuchis*.
- (2) Grazing in Registan all year was practised by some Baluch *kuchis* as long as drinking water and fodder were available.

In a report of the Food and Agricultural Organisation (FAO 2003) it was stated that some *kuchis* from Registan migrated across the Pakistan border for the summer months. None of the *kuchis* whom we interviewed mentioned this migration, however.

Drought of 1998–2002

In 1998, an extreme drought brought disaster to Afghanistan. Combined with many years of war, landmines and devastated livestock numbers it caused the death of many Afghans. The average number of sheep per family decreased from

21.9 to 2.9 (FAO 2003). Registan *kuchis*, in particular, were affected greatly. The Taliban, who were in power at the time, evacuated them with their remaining livestock to refugee camps at Maywand and Panjwaji near the Arghandab River, where they have remained until today (Frontier Post 2000). Other *kuchis* crossed into Pakistan where refugee camps were established. Three years of drought followed and over the entire period of drought over 90 percent of their sheep and about 40 percent of their camels perished. About 60 percent of *kuchi* households lost all livestock. The Taliban Government was unable to support these people and the UN has been providing food and clothing for the IDPs in the camps.

Four IDP camps were established near the Arghandab River in Kandahar province: one in the Maywand district (Qala-e Shamir) and three in the Panjwaji district (Marghar, Talukan and Moshan). *Kuchis* from these four camps were evacuees from Registan in the Kandahar province and the Garmser well cluster in Helmand province. Our study was directed towards the IDPs in these four camps. The number of animals owned per household in the IDP camps is generally very low. Three methods of raising sheep are presently being practised:

- (1) Zero grazing: seasonally available fodder, including straw, grasses and grape leaves, is collected from irrigated land near the river and transported back to the camp, usually by camel.
- (2) Aftermath grazing: animals graze on rented aftermaths of mainly wheat, but also other crops such as maize, for all or part of the year; zero grazing is practised when aftermaths are not available.
- (3) Migration to Registan: sheep from a number of families are combined into a flock of 100–300 head and shepherded by one or two families to Registan. Water is provided from wells or from tanks that are brought out by tractor. Sheep are maintained in Registan in winter/spring for as long as possible and then returned to IDP camps where each family provides for its sheep by one of the above two methods.

Payment for fodder collected or for land grazed can be in cash but usually is in exchange for work done by the sheep owner for the landowner.

Livestock

Kuchis raise mainly sheep but also have goats, camels and donkeys. As the restoration of sheep numbers is the main consideration, only sheep and their management will be described in detail in this report.

The only sheep breed raised by *kuchis* in Registan is the fat-tailed *Baluchi*. Like all fat-tailed sheep, they are well adapted to desert conditions, are good walkers and grazers, require relatively little water and are not prolific breeders. They are mated in summer during August, so that lambing coincides with the emergence of ephemerals after the first rains during January. Usually one ram per ten to twenty

females is used for mating. The rate of twinning is low, averaging about 5 percent of the ewes lambing. In the total flock, ninety to ninety-five lambs are obtained from one hundred ewes under a regime of good nutrition, but this can be reduced to only fifty lambs per hundred ewes under poor nutritional conditions. Until they are weaned, lamb mortality can range from 5 percent with good nutrition to 25 percent with poor nutrition. Lambs are weaned at five to six months of age and weigh 20–25 kg. The ram-lambs are sold in September/October for about 3,500 Afghanis (AFA: 50 AFA = 1 US\$), mainly to feedlot owners for fattening. Some ram-lambs are kept for a year or more until they weigh about 40–50 kg and can be sold for 8,000–10,000 AFA. The number and timing of the sale of ram-lambs depend on cash requirements and in practice, *kuchis* prefer to hold on to their sheep for as long as possible.

Ewes that are milked produce about 0.5 litres of milk per day and 100 litres of milk per lactation period. Milk is mainly for family consumption and some is processed into butter, ghee and dried yoghurt. Any surplus is sold. Sheep produce between 2.0 and 2.5 kg carpet wool per adult animal per year. One kilogram of white wool is sold for 50 AFA and one kilogram of black wool at AFA 15–20. Baluch *kuchis* eat old, dying sheep and sick lambs, slaughtering them just before death, but Pashtun *kuchis* do not.

Veterinary services are not used by the *kuchis*. Livestock are neither vaccinated nor dewormed. However, according to Dr Ali Khan, a VARA veterinarian, there is a veterinary clinic at Qala-e Shamir which services the four IDP camps. In addition, a mobile clinic visits the camps once a week. The most prevalent diseases are anthrax and enterotoxemia; brucellosis and foot and mouth are not problems. No vaccines are offered, but deworming is available for free once a year. The *kuchis* we interviewed were unaware of the above veterinary services and claimed that their livestock was never treated for parasites. They informed us that they purchase medicines for livestock, often in Pakistan, when the animals are ill.

Wells and Nawars

CADG, lead by Abdul Ali (Jan Agha) surveyed wells in the twelve clusters in Registan (see Fig. 1). Of seventy-six wells examined, eight were functioning, fifty-nine had collapsed and nine were dry. The eight functioning wells were located as follows: Gamay (well cluster No. 1) – one well; Jandozi (No. 2) – three wells; Choony (No. 6) – two wells; and Taghazi (No. 9) – two wells. VARA also surveyed about the same number of wells and found that six wells were functioning. The last well dug by *kuchis* was thirteen years ago at Jahanan (No. 11), while the UN dug one well at Jahanan (No. 11) and two wells at Taband (No. 12) five years ago. Wells collapsed or dried up following the three to four years of drought.

In general, wells are about 1–1.5 m square and 90–140 m deep, but can be as shallow as 21 m. They are hand dug by lowering a well digger with ropes secured

under the thighs, using camels to lower and raise the diggers. Typically, the upper part of the well (5–15 m) is of sandy soil that is naturally unstable and, therefore, reinforced with wood that requires maintenance every year or two, but can be reinforced with bricks and cement that require very little, if any, maintenance.

We visited the functioning well at Gamay (No. 1). It was 21 m deep to the water table, with slightly saline water drunk by both *kuchis* and livestock. We also visited two functioning wells at Taghazi (No. 9). These were 67 m deep to the water table with brick and cement reinforcement on top. They had been used recently and, apparently, *kuchis* had left three days before we arrived (on 24 May). Fresh sheep droppings were abundant around the wells and the rubber pails and ropes used to draw water were still there. Another well nearby with a wooden reinforcement had collapsed and a third well with wooden reinforcements had collapsed and was completely covered with sand.

A large number of *nawars* exist in Registan, their volume ranging from several cubic metres to thousands of cubic metres. Many of them are partially or completely clogged with sediments that migrated with floods or with eolic sand from nearby sand dunes. Since the *nawars* have not been maintained since 1998, their capacity to hold water has been reduced over the years. Although *nawars* appear to be a major source of water for *kuchis*, we saw only one large *nawar* (perhaps the largest in Registan), which was in disrepair, during our trip. This *nawar* was approximately 100 m long by 30 m wide. We were told that in good rainfall years the water was up to 3 m deep which is equivalent to about 9,000 m³ of water. Although the *nawar* was dry, the soil underneath was moist, indicating that there had been run-off water this year. It should be remembered that *nawars* are unreliable sources of water as they are totally dependent on seasonal rain. In drought years, *nawars* remain empty while wells still produce water. However, in rainy years, *nawars* could provide large volumes of high quality water.

Interviews at IDP Camp (Marghar): 1,800 Families

Interviews at IDP camps were difficult. It was nearly impossible to talk to one person at a time. As soon as an interview would start, other *kuchis* would gather around and often a consensus was taken among them to answer a question. It seemed that many answers that were provided were ones that the *kuchis* thought would best serve their interests. For example, if they thought that the purpose of our visit was to restore the number of sheep they had before the drought, they would most likely exaggerate the number of sheep. One *kuchi* claimed that he had 1,500 sheep before the drought; many others claimed that they had more than 500 head.

Interview with Khan Jan, chief of 350 Baluch families, comprising more than three thousand people

These families are from the Jandozi well cluster. Before the drought, there were about two thousand families at this cluster and over twelve thousand people. Each family had about four hundred sheep and, in all, there were ten thousand camels. They usually stayed at the well cluster for eight months and then migrated to river areas. Today, of the 350 families, only twenty-five to thirty families (not including those that have very few sheep) have a total of 1050 sheep, that is, ten to one-hundred sheep per family. This year, all 1050 sheep migrated to Jandozi in November for two months, returning to the IDP camp in January. For the other ten months in the camp, the families maintained their sheep either by zero grazing or grazing field aftermaths. They worked mainly for field owners as payment for fodder gathered or for grazing rights.

Interview with the chief of a tribe of four hundred families from the Attal Well cluster

Before the drought there were four to five thousand people and twelve thousand sheep at Attal. Some of these *kuchis* remained in Registan all year, while others migrated for part of the year. Today there are four to eight hundred sheep in total in the camp and only one of twenty wells at Attal is functioning. None of the families went to Registan this year.

Interviews with kuchis from Non-IDP settlements near the Arghandab river

These interviews were held with *kuchis* grazing their livestock near the rivers or at non-IDP settlements.

A kuchi grazing his sheep near the Helmand river

He owned one hundred ewes and two rams and claimed that all the sheep were needed for sustainability. Since the drought he had been renting land of wheat aftermaths for six months at 2000 AFA per month and buying straw for 16,000 AFA which he transported to Registan where he kept his flock for six months. He kept two mature rams for the flock at his settlement. On average, the one hundred sheep produced forty lambs: twenty males and twenty females. He sold ram-lambs as money was needed, usually in spring or autumn at a live-weight of 16–20 kg for 3,000 AFA per head, and kept ewe-lambs with the flock. Sheep were sold in Lashkar Gah or Kandahar to traders, butchers etc. Ewes were milked until spring for home consumption. On the day of the interview, he obtained 10 kg of milk from one hundred ewes. His sheep produced 2.0–2.5 kg wool per sheep, which was currently being sold at 25 AFA per kg. He slaughtered and ate fifteen to twenty old and young, ill sheep just prior to their death. No veterinary service was offered to his animals; occasionally he would purchase drugs if sheep were ill.

A kuchi interviewed in a settlement at the edge of Registan:

He owned three of thirty-two wells in a cluster, but none of his wells was functioning. He claimed that he had deeds to these wells. Before the drought, there were another fifteen families at his wells and water was given free of charge. At that time, he had five hundred sheep while the other families had about fifty sheep each. They used to stay in Registan all year as there used to be enough pasture. Today he has twenty sheep and rents fields near the Helmand river for eight months for 4,000 AFA; the rest of the year he grazes his sheep in Registan.

Another kuchi in a settlement at the edge of Registan:

Before the drought, he had five hundred sheep which produced about four hundred lambs in Registan. Today he has thirty sheep and practises zero grazing by bringing fodder (grasses, wheat straw) to his sheep by camel. He worked as a labourer in fields of farmers in exchange for fodder. The thirty sheep produced ten to fifteen lambs which were sold at six months of age at 10–25 kg live-weight for 3,500 AFA each. Ewes were milked and, at peak lactation, eleven ewes produced 8 kg milk. He claimed that sheep and lambs were healthier in Registan than in the settlements. No veterinary services were offered to the livestock.

A kuchi interviewed from the Taghazi well cluster:

He owned 250 ewes and three rams before the drought; today he has seven sheep. Before the drought, he stayed in Registan during the winter and then migrated north to the river area. From 250 sheep, about 160 lambs were produced, of which 30 were eaten and, therefore, 130 lambs (about 65 each of males and females), were left. About ten ewes died annually. He held ram-lambs for two years, at 20–25 kg live-weight, then sold them for 3,500 AFA each. He milked ewes for milk and milk products (cheese, ghee, dried yoghurt) for family use, but had surplus to sell. White wool was sold for 50 AFA and black wool for 15–20 AFA per kg. Before the drought, he also had twenty camels, which were used for transport, drawing water and as a source of cash. Each camel was worth about AFA 25,000, about the same price as today; now he has none.

Another kuchi from the Taghazi well cluster:

He was head of his tribe and owned four wells; three were brick- and cement-reinforced and still functioning and one was wood-reinforced and had collapsed. He had deeds to the wells. He was about sixty years old and the wells were about seventy years old, originally built by his father. Water from the wells was suitable for humans. Before the drought, he and his nine brothers had 300–400 sheep and 25–30 camels and he usually stayed at one of the wells. In all, there were 3,000–4,000 sheep and 500–600 camels at the three wells. Sheep were watered during the day; camels at night. He stayed at the wells for eight months and then migrated to near the Helmand River for four months. Today, he and his brothers

have seventy sheep and five camels. This year and throughout drought years, he migrated to Registan for one or two months and stayed near the Helmand River for ten or eleven months. Fodder and water were not a problem; he used aftermaths from wheat, cotton and corn fields. He either grazed his animals or brought back fodder by camels. He worked for the field owner in exchange for the fodder. From his seventy sheep, forty to fifty lambs were produced. Weaned males were sold, at five to six months of age and weighing 10–15 kg, for 2,500 AFA each. All females were kept. He claimed that it was rare for one person to own large flocks of five hundred or more sheep; such large flocks were usually owned by a number of persons together, as was the case with him and his brothers.

Discussion

From a review of the reports and articles published by NGOs and UN agencies and our observations and interviews, the following discussion indicates what the situation of the Baluch and Pashtun *kuchis* was in the southeast Registan region of the Kandahar and Helmand provinces before the onset in 1998 of the four years of drought.

Population

In all there are approximately 9,600 families in the four IDP camps or about 57,600 people (assuming six persons per family, according to UNHCR), which is about 60 percent of the families that inhabited the above area. The other 40 percent are living in unsupported, spontaneous settlements mainly situated between the Arghandab and Helmand Rivers and Registan, but not in IDP camps. Thus, in all, there are about 16,000 displaced families or 96,000 people. Many of the men have become farm labourers in areas such as Helmand and Kandahar where crop irrigation is practised (UNCHA 2004). The great majority of these *kuchis* are illiterate, so that apart from the census carried out in Afghanistan on a district basis in 1995, 1998 and 2003 (FAO 2003), there are no verifiable vital statistics available. Even the 2003 census is equivocal, because there is no information available to distinguish the *kuchi* inhabitants of Kandahar Registan from the general population of the Kandahar and Helmand provinces.

Table 1: Numbers of Families in IDP Camps, 2003 (Van Engelen 2003a)

Camp	Sections	Families
Marghar	9	1,957
Qala-e-Shamir	10	1,440
Talukan	27	5,225
Moshan	18	3,470
Total	64	12,092

Table 1 shows the results of a family count carried out during 2003 by VARA. The total number of families is 26 percent greater than the 9,600 families quoted by UNHCR. We are unaware of the reason for this significant discrepancy, as the methodology used for the family count by VARA was not discussed. We assumed that the UNHCR count is likely to be more reliable only because we presume that all families in the camp would seek subsistence aid from the UN and its relevant agencies. Heads and/or chiefs of the various sections in the camps would tend to exaggerate rather than lower the number of families in their section in order to receive more humanitarian aid.

It should be noted that the population of the IDP camps remains equivocal. For instance, van Engelen (2003b) reported that in a tent-to-tent survey in the camps carried out for VARA, the UNHCR/WFP used a population estimate that was 33 percent higher than what was found.

Area of Registan and its Climate

The estimated area of Registan that was utilised by the 16,000 families was approximately 10,000 km². The area is considered to be a desert, having an arid climate characterised by low Mediterranean type rainfall, high summer ambient temperatures and relatively cold continental winters. Because there are no meteorological records from Registan, the annual rainfall can only be estimated from records of meteorological stations at Lashkar Gah and Kandahar towns. The most often quoted estimate is an average annual rainfall of up to 90 mm. The ambient temperatures recorded at Kandahar are summer highs of well over 40°C and winter lows of slightly above zero. Although *kuchis* cannot quantify rainfall, they do claim that they had never experienced a drought as severe as that of 1998, nor experienced continuous drought years like those between 1998 and 2002. Although this was never stated by *kuchis* we interviewed, such an arid climate indicates an ecosystem subject to fluctuations within and between years in terms of soil water and fodder availability that would necessitate nomadic practices for survival.

Soils and Vegetation

According to recent soil surveys carried out independently by Kristina Toderich (2004) and Abdul Ghani Ayubi (2004) for CADG, ‘vast areas are covered by immature soils derived from unconsolidated materials, such as gravel, rocks and sand’ (Toderich, 2004: 4). They found almost all the varieties of geomorphic and desert type soils: hammadas, regs, ergs, *nebkhas*, *takers* and playas (Afghan terms in italics). In layman terms, the most important characteristics are the presence of mobile sand dunes, separated by hammadas (soils covered with a top layer of small stones, with varying slopes) that protect relatively stable sandy and loess

soils. Hammadas are important as being relatively free of vegetation and are conducive to water run-off for potential water collection when there are rainfall events of sufficient intensity. Sandy and loess type soils are those that support the more important (for livestock) desert-type vegetation. For this discussion, the most important vegetation types are perennial and annual *psammophytic* species, and annual ephemeral vegetation.

Perennial and annual *psammophytic* species predominate on sand dunes and sandy fixed areas. Tree and shrub perennials are relatively unpalatable to sheep but more palatable to camels and donkeys. Perennials are important for protecting sandy fixed areas and well clusters from sand encroachment, for stabilising sand dunes, for shade, for fuel and for *kuchi* dwellings and well construction. During the drought, many shrubs were collected by *kuchis*, who travel into Registan by tractor, and the shrubs were sold for firewood in Lashkar Gah and other centres.

Annual ephemeral vegetation predominate on sandy-loams and loess soils. They are highly palatable and constitute the main fodder intake of small ruminants grazing in Registan. This vegetation usually and universally rejuvenates following a significant rainfall event, even after a drought year and particularly after continuous drought years. The reason is because the soil seed-bank is replenished in the absence of a grazing predator of the few plants that do germinate and seed. For more detailed and theoretical explanations, see Rosenzweig and MacArthur (1963) and Noy-Meir (1996).

The sustainable annual carrying capacity of Registan was that of the total production system, that is, the number of animals that was maintained by *kuchis*, not only by grazing in Registan but also by migrating to other areas. In these terms it is best to refer to the length of time spent in Registan by *kuchis* as a point of reference. Time spent in Registan in any particular year was determined by soil water availability for plant growth (edible biomass) and drinking water available from wells and *nawars*. Interdependent relationships between vegetation and drinking water availabilities for livestock determined the transhumance of *kuchis* from Registan to the semi-arid and irrigated regions adjacent to the Helmand and Arghandab rivers.

Livestock

A reliable estimate of the number of animals held by the 16,000 *kuchi* families before 1998 is not available from the 1995 census conducted by the FAO for the same reasons as discussed above for determining population numbers. The statistics most often quoted by the various investigating agencies were that 90 percent and 60 percent of the sheep and camel populations, respectively, perished following the disastrous droughts of 1998. Any estimate is thus dependent on the estimate of present sheep and camel populations. Table 2 shows the results of an animal count carried out in the four IDP camps during 2003 by van Engelen (2003b).

Table 2: *Estimated Sheep and Camel Populations in IDP Camps, 2003 (Van Engelen 2003b)*

Camp	Sections	Families	Camels	Sheep/Goats
Marghar	9	1,957	NA	NA
Qala-e-Shamir	10	1,440	996	4,693
Talukan	27	5,225	NA	NA
Moshan	18	3,470	981	10,289
Totals counted	64	12,092	1,977	14,982
<i>Animals calculated per family</i>			<i>0.40</i>	<i>3.05</i>
Estimated total animals in camps			4,837	36,881

However, if we assume that the animal head count is correct, but that there are 26 percent fewer families in the IDP camps (as discussed above), there could be 26 percent more camels and sheep per family. On this basis there are perhaps 0.050 camels and 3.84 sheep/goats per family. Thus the estimated total animal numbers could have been approximately five camels and thirty-five sheep/goats per family before 1998.

In a survey of ten representative *kuchi* families out of 6,400 that once lived at different well clusters in Registan but are now living in unsupported spontaneous settlements, Ayubi (2004) reported that on average they owned thirty-one sheep and one camel per family. According to these families, before the drought they owned approximately 263 sheep and 21 camels per family. The existing numbers of sheep and camels per family are 11.8 percent and 4.8 percent, respectively, of the sheep and camels they previously possessed. While the percentage of remaining sheep per family is relatively consistent with that generally accepted, the remaining camels are very much less than the 40 percent that is commonly quoted. The reason for this discrepancy is unclear. Possibly more camels survived the drought but were subsequently sold for any number of reasons. What is of importance from Ayubi's report is that families possessing on average thirty-one sheep and one camel were subsisting unsupported.

Water Resources

In the southeastern Registan, there are twelve well clusters spread unevenly over the area. According to the recent survey undertaken by VARA to map these clusters, 380 wells have been identified and mapped (van Engelen 2003a). Of these identified wells, few were functioning and water was drawn from them from depths as great as 140 m (a consequence of low water tables). In general, the functioning wells were those that were brick-lined in their upper reaches, with surfaces protected by concrete. According to van Engelen (2003b: 2), there are

30–40 traditional well diggers/maintenance specialists in the IDP camps who are currently unemployed.

No research has been made to explore the aquifer unit underlying the different well clusters in Registan. Therefore, the potential of each well to provide more water, the type of aquifer, its hydraulic properties and its response to droughts are unknown. Local hand-dug wells penetrate only a few metres below the water table (due to technical limitations) and the volume of water pumped from each well is very small (limited by the time that the bucket takes to be lowered into the well). In addition, there are *nawars* associated with these well clusters. These are depressions where run-off water is collected from sloping hammada areas when rainfall is of sufficient intensity to cause run-off. After 4–5 years of neglect, most of these *nawars*, like most of the wells, have been filled and covered with sand. It is probable that the *nawars* were conveniently utilised by grazing animals until depleted and only then was water labour-intensively drawn for the animals from wells.

There are no formal owners of wells, although some *kuchi* chiefs interviewed claimed they had deeds of ownership. However, it appears that a well cluster was controlled by a group of related families by virtue of what had become their traditional occupancy of the pasture areas surrounding the well cluster. Probably the extent of control may have been a radius of up to 10–15 km from a well cluster, a shepherding distance that could allow easy watering of livestock in the presence of water in *nawars* but would be more difficult in their absence or after their depletion. Traditionally, it would appear that it was customary to allow families from other groups to draw and transport water to their animals but they could not bring their animals to within walking distance of the wells. In this way pasture and browse resources were reserved for livestock of resident families.

Water and Pasture Inter-Relationships

In any particular year, the length of time spent in the Registan region before eventual migration was restricted either by pasture availability or the drinking water available and possibly, but rarely, by both (when dependent on *nawars*). However, when wells were functioning, it is probable that pasture availability was the main constraint limiting the time spent in Registan. The beginning of the time spent in Registan would have had to be when vegetative growth, following a significant rainfall event(s), had progressed sufficiently to allow animals to forage enough fodder to approach their maintenance requirements. Because the number of animals foraging was a function of the stocking rate of the year-round nomadic system and not the actual carrying capacity of Registan fodder resources, overgrazing would have been the rule rather than the exception. That is, grazing/browsing continued until inevitably all edible biomass had been consumed after a few months. If foraging began too early, animal intake rates would be

greater than plant growth rates which would lead to an early migration. The initiation of foraging would have had to take into consideration the availability (purchase and/or renting) of fodder resources on the migratory route to the semi-arid and irrigated regions adjacent to the rivers. This reasoning was reinforced by interviewed *kuchis* who, each having had access to a functioning well, had returned during the winter/spring seasons of 2002/2003 and 2003/2004 to Registan. They reported that even though their flocks were only a tenth of their former size, they had returned to the IDP camps after only two to three months. Possibly, however, they may have also returned because of work prospects in harvesting wheat and aftermath availability at the irrigated areas.

Socio-Economic Factors

It would be a mistake to underestimate the indigenous knowledge of the ecosystem that the *kuchis* had utilised for centuries without outside interference. As previously mentioned, there were (and are) specialists among them whose responsibility it was to dig and maintain wells and *nawars*. The shepherds were also knowledgeable and experienced in the use of the grazing and browsing resources traditionally available to them in Registan. It is naive to think that they had never experienced the debilitating effects of drought conditions. Under such conditions, they migrated earlier to where fodder was traditionally known to be available. Only two options would be available to them: either the preferred choice of bringing animals to fodder or bringing fodder to the animals. In the absence or partial absence of these options due to fodder supply not being sufficient to meet demand, animal numbers were reduced by mortality or slaughter and sale according to subsistence needs. Following a drought, restoring the number of animals formerly held was a function of the fertility rate of the remaining breeding animals.

It would appear that *kuchis* practised a well defined transhumance form of nomadism in which they always returned during the winter/spring seasons to grazing/browsing areas that were centred around maintained wells and *nawars* to which they had traditional rights of use. The dramatic change caused by the disastrous years of drought and the evacuation of *kuchis* with few animals to IDP camps is that the IDP camps have replaced their traditional areas in Registan as their home territory to which they return if they migrate. This is because few of the wells and few, if any, of the *nawars* are functioning in Registan, whereas at the IDP camps, water and humanitarian aid is readily available at no charge. In addition, for those *kuchis* with animals, fodder is available albeit at a price. Self-interest guides them to exploit the resources currently available to them.

The 1998 evacuation to IDP camps of Registan *kuchis* by the then ruling Taliban Government was for economic reasons. *Kuchis* had lost so many of their animals that their means of livelihood no longer existed and their main means of

livelihood became the humanitarian aid provided by outside agencies, mainly the UN. Unlike in many other war-affected countries, where refugee camps have become and are becoming permanent fixtures maintained by aid agencies, *kuchis* are not prevented from returning to their former unoccupied territories by threats of violence from opposing forces, but remain only because of the lack of economic incentives. Because they do not have the earning capacity to purchase sufficient numbers of animals needed for their traditional form of livelihood and in the absence of any other means, they are fated, if only by self-interest, to remain in IDP camps as long as humanitarian aid is forthcoming. Unless the present conditions of self-interest for *kuchis* are changed, the IDP camps are liable to become a self-perpetuating and permanent financial burden on a reluctant UN. For the UN, the solution in the short term is to continue to fund the camps for humanitarian reasons and to fund consulting NGOs in an endeavour to find long-term solutions to the problem of how to provide the economic incentives for *kuchis* to return to their traditional way of livelihood.

Notes

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1. Countries bordering Afghanistan include Iran to the west (936 km border); Turkmenistan to the northwest (744 km); Uzbekistan to the north (137 km); Tajikistan to the northeast (1,206 km); China to the eastnorth (76 km) and Pakistan to the east and south (2,430 km).
2. The Registan desert's geographical boundaries lie roughly between 29°30' and 30°45' north and 64°25' and 55°05' east (CIA 2004).
3. Our group included an organiser, field manager and interpreter from CADG, *kuchi* guides and security personnel.

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